

CRESS for Auditing Few-Shot Anomaly Reliability: Resolution Floors and Cross-Category Certification Feasibility

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ABSTRACT

Few-shot industrial anomaly detectors that utilize frozen foundation features often achieve strong anomaly rankings, but their thresholds may not maintain consistent false-alarm rates across different categories or under input distribution shifts. To address this gap, we introduce an auditable reliability framework with two complementary strengths: it exposes the exact resolution limit of target-only alarms and tests whether source-category evidence is sufficient for transferable certification, rather than presuming that source pooling ensures control. The framework integrates a low-storage DINOv2 PCA residual with label-free leave-one-image-out (LOIO) p-values derived from k normal supports. Evaluation across 15 MVTec and 12 VisA categories under four corruption types reveals a resolution floor: target-only alarms cannot trigger below $\alpha = 1/(k+1)$. Empirical normal p-values are observed to be conservative on VisA at $k = 4$ and anti-conservative on corrupted MVTec. Additionally, Cross-category Reliability Estimation with Source Support (CRESS) is introduced as a source-assisted procedure that distinguishes reference, threshold-proposal, and certification categories. Across a frozen grid of four k values, five seeds, five conditions, within-dataset tests, and MVTec-to-VisA/MPDD transfer, strict category-level certification consistently returns the fail-closed zero threshold in all 960 gate cells. This result is not solely attributable to Hoeffding looseness: when all observed category losses are zero, even an optimistic, multiplicity-free, deterministic, uniformly valid, distribution-free 95% upper bound requires at least 14 independent categories to certify $\alpha = 0.20$. The frozen protocol requires more, whereas only three or four certification categories are available. Fixed-archive image certificates select positive thresholds in 37% to 60% of settings, but these estimate image-level risk under the observed source mixture rather than for new-category risk. Historical non-independent CRESS results are therefore retained solely as empirical diagnostics. Collectively, these findings establish a clear feasibility boundary: the framework distinguishes empirically useful source thresholds from transferable certificates and prevents unsupported claims of target-category control.

1. Introduction

Industrial anomaly detection (AD) is commonly implemented in asymmetric data regimes, where normal examples are available, while abnormal examples are rare, heterogeneous, and expensive to annotate. Consequently, few-shot AD has become practically significant. In this context, an inspection system typically receives only k normal support images for a target category, often with $k \in \{1, 2, 4, 8\}$, and is required to detect both image-level and pixel-level defects during testing.

Recent approaches have improved performance in this regime by leveraging frozen visual foundation models. For example, DINOv2 [1] patch features can be stored in a normal memory bank and queried using nearest-neighbor distances, as implemented in PatchCore-style methods and AnomalyDINO [2, 3]. Alternatively, frozen patch features may be compressed into a normal principal component analysis (PCA) or subspace model, as in SubspaceAD [4]. These methods enable effective anomaly ranking without the need to train a large task-specific backbone.

However, effective ranking does not equate to reliable decision support. A raw nearest-neighbor distance or PCA residual may yield high area under the receiver operating

characteristic curve (AUROC) while remaining poorly calibrated: a score threshold that is effective for one class may not correspond to the same false-alarm rate for another class, dataset, or under data corruption. This distinction is critical in industrial deployment, where operators require not only identification of suspicious samples but also trustworthy alarm probabilities. Previous studies have demonstrated that DINOv2-based few-shot AD can exhibit poor calibration and adversarial fragility [5]. As such, reliability is an integral aspect of the deployment challenge, rather than a superficial post-processing concern.

This work addresses the identified gap by decoupling anomaly ranking from reliability estimation. The central questions pertain to the learning system itself: what false-alarm behavior can a decision rule, calibrated on k examples of a frozen representation, guarantee, at what resolution, and how does this guarantee degrade under input distribution shifts? The ranking component utilizes a frozen DINOv2 PCA/subspace detector, aggregating patch residuals into a raw anomaly score for AUROC, average precision (AP), and localization. A distinct reliability component transforms this evidence into calibrated or probability-like outputs. The primary target-only approach employs leave-one-image-out (LOIO) conformal calibration on the limited normal support images. Each support image is evaluated as if held out,

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generating normal nonconformity values; a test image subsequently receives a conformal p-value and a reliability score of $1 - p$. This framework does not constitute a supervised anomaly posterior, but it is statistically interpretable and constructed without target anomaly labels.

The foundational components utilized in this study, DINOv2 features, PCA residuals, Platt-family calibration, and conformal anomaly detection, are well-established in the literature; the present contribution builds upon these elements. However, existing research leaves unresolved structural questions specific to the few-shot regime: for example, a conformal alarm calibrated on k support images cannot trigger below $\alpha = 1/(k+1)$, corruption shift can render the observed false-alarm rate anti-conservative, and a statistically valid source-category certificate may be vacuous when only a limited number of categories are available. The primary contribution integrates an exact resolution-floor diagnosis, an empirical audit under distribution shift, and **Cross-category Reliability Estimation with Source Support (CRESS)**, a source-assisted thresholding and certification procedure evaluated using a strict nested protocol. The rigorous experimental protocol does not support the anticipated new-category claim: all category-certified thresholds revert to zero. This negative outcome is consistent with both the stated Hoeffding rule and a broader all-zero lower bound for deterministic, distribution-free category-risk confidence bounds, and is reported alongside, rather than in place of, fixed-archive image sensitivity and historical empirical results.

The implications for learning systems are broadly applicable, despite the industrial testbed: finite calibration constrains both the resolution of target-only alarms and the feasibility of source-category certification. This paper presents four contributions, with the first two being central:

- We make the few-shot resolution limit of conformal alarms explicit through an attainable-alpha analysis: with k supports, no alarm can trigger below $\alpha = 1/(k+1)$, and empirical false-alarm rates must be interpreted relative to the attainable grid rather than the nominal level. Cluster-aware Monte Carlo diagnostics of normal p-values then localize when and in which direction the exchangeable-null pattern fails under corruption shift, being conservative on VisA and anti-conservative on corrupted MVTec.
- We formulate CRESS with disjoint reference, proposal, and certification categories and evaluate it under a strict nested protocol. In addition to the Hoeffding-specific feasibility condition, we prove that any deterministic uniformly valid distribution-free upper bound based solely on independent and identically distributed (iid) category losses has an all-zero floor $1 - \beta^{1/n}$; even without multiplicity, 95% certification at $\alpha = 0.20$ requires 14 categories. The frozen experiment spans four k , five random seeds, five conditions, MVTec, VisA, and MVTec-to-VisA/MPDD transfer. Category certification yields a

zero threshold in all 960 gate cells because only three or four certification categories are available; image-level fixed-archive sensitivity is reported separately and never relabeled as a new-category result.

- As supporting evidence, we evaluate the fixed low-storage ranker and compare LOIO with Vector/Shift-Aware and standard scalar calibrators under a common label-free protocol. These comparisons never alter the raw ranking, and operational metrics rather than ECE carry the central interpretation.
- Secondary diagnostics cover storage, transfer calibration, pixel-level localization, prevalence sensitivity, source-view selection, and adversarial fragility. They delimit the claim: the framework is not an AUROC state of the art or an adversarially robust detector, and historical non-independent CRESS results are not certificates.

The remainder of this paper is structured as follows. Section 2 positions the work against few-shot detectors, calibration, and conformal anomaly detection. Section 3 defines the frozen ranking substrate, target-only LOIO route, source-score construction, and nested CRESS certificate. Section 4 outlines datasets, protocols, and claim rules. Section 5 proceeds from ranking and target-only operation to the shift audit, strict CRESS result, and secondary diagnostics. Section 6 discusses limitations, and Section 7 concludes.

2. Related Work

2.1. Few-Shot Industrial Anomaly Detection

Classical industrial anomaly detection (AD) typically assumes access to normal-only training data and identifies deviations at test time. Patch distribution, self-supervised, and reconstruction-discriminative approaches such as PaDiM, CutPaste, and DRAEM provide important context for this literature [6–8]. PatchCore [2] established memory-bank patch retrieval as a robust baseline by storing representative normal features and scoring test patches using nearest-neighbor distance. AnomalyDINO [3] extends this paradigm to few-shot inspection with frozen DINOv2 representations. The few-shot AD literature has further introduced registration-based adaptation (RegAD [9]), graph-structured memory (GraphCore [10]), fast feature reconstruction (FastRecon [11]), vision-language zero- and few-shot scoring (WinCLIP [12]; CLIP-SAM collaboration [13] and cross-modal prompt regularization [14] continue this direction), and in-context residual learning (InCTRL [15]). While these methods are effective rankers, the reference state can remain nontrivial at inference time, and the raw score scale does not represent a calibrated probability. Calibration under distribution shift is also known to degrade for deep models generally [16]. Our work is complementary: it does not compete on ranking but instead examines how few-shot detectors should expose decision reliability.

2.2. Frozen-Feature Subspace Detectors

Subspace approaches replace explicit patch retrieval with a compact normal subspace. SubspaceAD [4] demonstrates that frozen DINOv2 patch features and PCA or subspace residuals are already effective for few-shot anomaly ranking, establishing the ranking mechanism as settled prior art. Our contribution begins after the residual is computed, focusing on how a low-storage subspace detector should expose reliability under low-shot and distribution shift conditions, as well as the guarantees its alarms can provide.

2.3. Calibration and Robustness in Few-Shot AD

Calibration evaluates whether predicted confidence aligns with empirical correctness [17, 18]. Standard post-hoc calibrators include Platt scaling [17], temperature scaling [18], histogram binning [19], and isotonic regression [20]; all are compared under a shared label-free protocol. In anomaly detection, calibration is challenging due to sparse labels, open-ended anomaly types, and varying score distributions across classes. Reliability analyses of DINOv2-based few-shot AD report poor expected calibration error (ECE) and significant FGSM fragility [5]. These findings motivated us to evaluate ECE, Brier score, and negative log-likelihood (NLL) separately from AUROC and average precision (AP), and to report adversarial results, including FGSM-style perturbations [21], only as fragility diagnostics rather than as evidence of improved robustness.

2.4. Conformal Prediction and Risk-Controlled Calibration

Conformal prediction transforms nonconformity scores into p-values under exchangeability assumptions [22–24]. This approach is particularly suitable for normal-only few-shot AD, as support normal images can define a reference distribution without requiring real anomalies. Conformal anomaly detection was already well established: leave-one-out, bootstrap, and cross-conformal anomaly detectors had been directly analyzed [25]; weighted conformal detection in low-data regimes had examined resolution and effective-sample-size effects [26]; general nonconformity toolkits had been developed [27]; and few-shot conformal prediction with auxiliary tasks had addressed small target calibration sets [28]. Conformal AD, LOIO calibration, auxiliary-task transfer, and weighted conformal p-values are therefore established mechanisms.

Risk-controlling calibration also preceded our work. Learn then Test cast finite-sample calibration as a multiple-testing problem over candidate configurations [29], while conformal risk control selected a parameter to control the expectation of a monotone loss [30]. In anomaly detection, PAC-Wrap provided semi-supervised PAC bounds for false-positive and false-negative rates around a base detector [31]. These works established the general principles of candidate testing, held-out risk assessment, and one-sided error bounds; we do not claim those statistical primitives as new. CRESS instead instantiates them for normal-only cross-category thresholding by assigning disjoint source

categories to reference, proposal, and certification roles, distinguishing fixed-archive image risk from new-category risk, and auditing whether the available number of independent categories can support a nonzero threshold.

Structure-aware conformal adaptation had also been studied outside anomaly detection. Lam and Anh proposed HeAD-CP, which adapted graph-score diffusion using label-free local-homophily estimates and showed that uniform diffusion could impair prediction-set efficiency on heterophilic graphs while preserving marginal coverage [32]. This result supports the broader need to audit reliability transformations under structural shift, but its setting concerns GNN node-classification sets rather than few-shot industrial anomaly alarms or cross-category source certification.

The present contribution begins at the few-shot resolution floor $1/(k+1)$ below which no deterministic target-only k -score conformal alarm can operate, and at the independent-category requirement of a simultaneous source certificate. This work establishes the algebraic floor and the Hoeffding-specific and distribution-free all-zero category-feasibility bounds, empirically audits corruption-shift deviations, and evaluates source-assisted candidate thresholds without target anomaly labels. CRESS does not introduce a new nonconformity score or generic confidence bound; it specifies how source categories are separated into reference, proposal, and certification roles and audits whether the resulting source certificate can be nonvacuous at the available category count.

3. Method

3.1. Problem Setup

For each target category c , we observe a support set $S_c = \{x_i\}_{i=1}^k$ containing only normal images. Target test images may be normal or anomalous, but their labels are reserved exclusively for evaluation and do not enter detector fitting, threshold proposal, or certification. Other categories may additionally provide a declared source-normal archive. The framework produces a fixed raw anomaly score $s_c(x)$ and optional pixel map, a target-only reliability value $p_{\text{LOIO}}(x)$ when leave-one-image-out calibration is available, and, for the source-assisted route, a threshold τ^* accompanied by a source-domain certificate. Figure 1 separates the inherited ranking substrate from these two reliability routes.

3.2. Frozen Patch Features and Subspace Ranking

We extract patch features with a frozen DINOv2 ViT-S/14 backbone. Let $z_{ij} \in \mathbb{R}^d$ denote patch j from image i . A PCA subspace is fit on the target support patches. With support mean μ_c and retained principal directions $U_c \in \mathbb{R}^{d \times m}$, the patch residual is

$$r_c(z) = \left\| (z - \mu_c) - U_c U_c^\top (z - \mu_c) \right\|_2. \quad (1)$$

The patch residuals form an anomaly map. The image-level ranking score is

$$s_c(x) = \text{Agg}_j r_c(z_j). \quad (2)$$

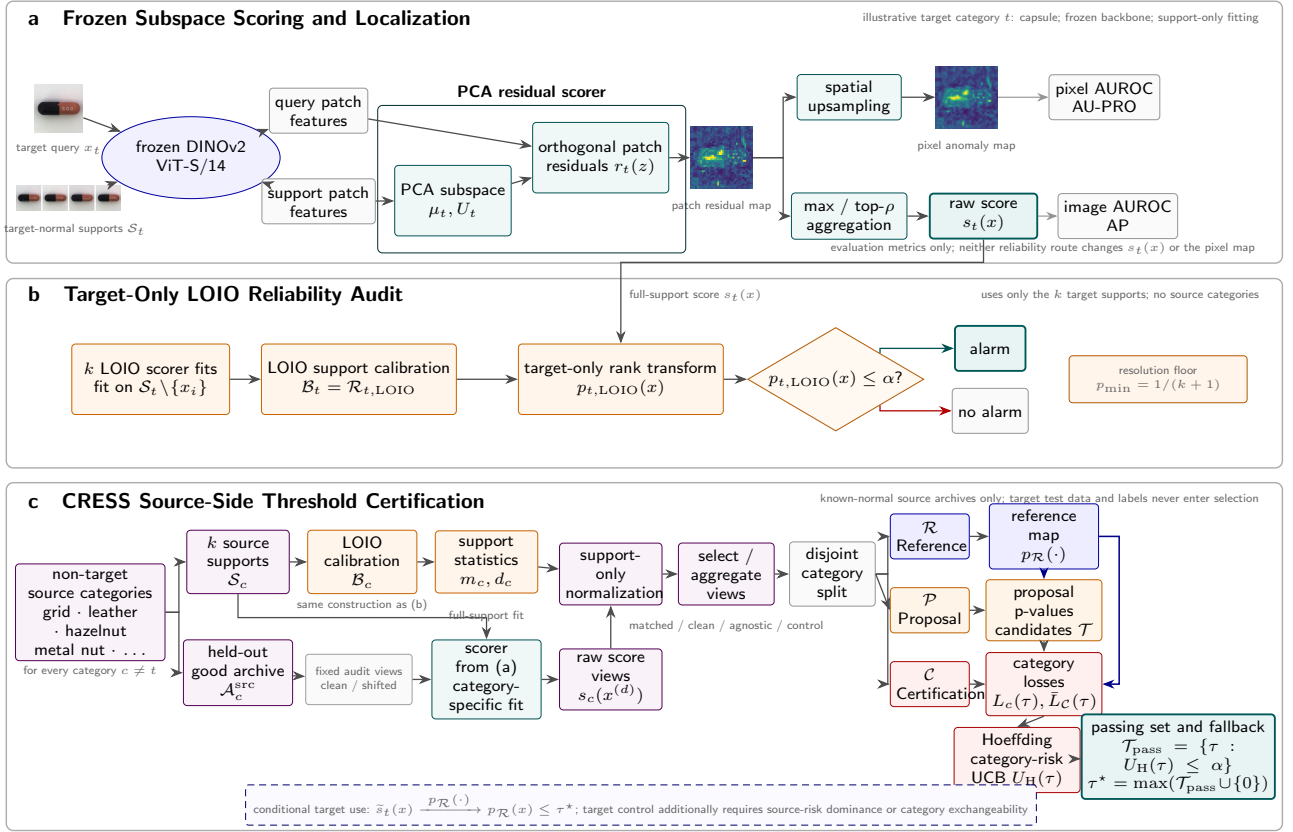


Figure 1: Overview of the audited reliability framework. Panel (a) shows the shared frozen DINOv2 PCA-residual scorer, illustrated with a traceable capsule example, and its fixed image-ranking and localization outputs; the metric boxes are evaluation-only. Panel (b) applies the same scoring construction in leave-one-image-out folds to obtain target-only calibration residuals and exposes the smallest attainable p-value $1/(k+1)$. Panel (c) is the separate source-assisted CRESS route. For every non-target source category, its supports provide both the category-specific PCA fit and LOIO-derived normalization statistics, while a disjoint held-out good archive provides clean or fixed audit-corruption score views. After support-only normalization and condition-view selection, source categories are split into disjoint reference, proposal, and certification roles: $\mathcal{R}$ defines $p_{\mathcal{R}}(\cdot)$, $\mathcal{P}$ yields proposal p-values and candidate thresholds, and $\mathcal{C}$ yields category losses for the family-adjusted Hoeffding category-risk UCB. CRESS returns the largest passing candidate or zero; the final strip states the additional assumption required to transfer the source-domain certificate to the target category.

The clean accuracy-storage benchmark uses the patch maximum, whereas the conformal protocols use the mean of the top-1% patch residuals ($\rho = 0.01$) as a smoothed maximum. Section 5 reports an aggregation-sensitivity ablation. AUROC and AP are always computed from the fixed score $s_c(x)$; none of the reliability transformations changes this ranking.

3.3. Target-Only LOIO Reliability

For $k \geq 2$, the target-only route constructs normal calibration residuals by leave-one-image-out support splitting. For each support image x_i , a subspace is fit on $S_c \setminus \{x_i\}$, and the held-out image receives the score $s_{c,-i}(x_i)$. The resulting set

$$\mathcal{R}_{c, \text{LOIO}} = \{s_{c,-i}(x_i) : x_i \in S_c\} \quad (3)$$

serves as target-normal nonconformity evidence. A test image, scored using the subspace fit on all k supports, receives

$$p_{c, \text{LOIO}}(x) = \frac{1 + \sum_{r \in \mathcal{R}_{c, \text{LOIO}}} \mathbf{1}\{r \geq s_c(x)\}}{1 + |\mathcal{R}_{c, \text{LOIO}}|}. \quad (4)$$

We report $1 - p_{c, \text{LOIO}}(x)$ only as a probability-like anomaly reliability view. It measures extremeness relative to held-out support normals and is not a supervised anomaly posterior. Weighted conformal replaces the uniform count with density-ratio weights [33] and is evaluated as a secondary baseline.

The construction is asymmetric: each calibration score uses $k - 1$ support images, whereas the test score uses all k . Equation (4) is therefore a LOIO approximation in the spirit of cross-conformal inference [22, 25, 34], not a split-conformal p-value with an automatic finite-sample guarantee [35, 36]. A matched-LOIO audit scores held-out support and test images under the same fold-specific subspaces, but does not improve the observed false-alarm and power trade-off. We consequently retain the simpler construction and

audit its normal p-values empirically rather than assuming exchangeability.

3.4. Attainable Alpha and the Few-Shot Resolution Floor

Proposition 1 (attainable-alpha floor). For any finite calibration scores $r_1, \dots, r_k$, a p-value of the form in Eq. (4) belongs to $\{j/(k+1) : j = 1, \dots, k+1\}$. Hence the alarm $\mathbf{1}\{p(x) \leq \alpha\}$ is identically zero for $\alpha < 1/(k+1)$.

Proof. The exceedance count is an integer from 0 to k . Adding one and dividing by $k+1$ gives the stated grid and lower bound. No exchangeability assumption is needed for this algebraic claim. $\square$

For $k = 4$, the floor is 0.2; for $k = 8$, it is $1/9 \approx 0.111$. The attainable-alpha analysis reports, for each nominal α , the largest attainable grid point not exceeding α and the corresponding empirical false-alarm rate. The floor is a finite-resolution property, not an algorithmic innovation or evidence that the underlying ranker cannot separate anomalies. Smoothed randomized p-values provide a standard way to remove the deterministic floor under the relevant exchangeability assumptions; they are retained as a baseline because randomization does not itself repair support-to-test shift.

3.5. Source Score Construction and View Selection

CRESS uses known-normal images from categories other than the target. For every target or source category c , let $\mathcal{B}_c$ denote its support-only calibration residuals. For $k \geq 2$, $\mathcal{B}_c = \mathcal{R}_{c, \text{LOIO}}$. Because LOIO is undefined with one support image, the declared $k = 1$ stress test uses one patch-split support calibration residual instead. Let

$$\begin{aligned} m_c &= \text{median}(\mathcal{B}_c), \\ d_c &= \text{MAD}(\mathcal{B}_c), \\ \tilde{s}_c(x) &= \frac{s_c(x) - m_c}{\max(d_c, \varepsilon)}. \end{aligned} \quad (5)$$

where $\varepsilon = 10^{-6}$ is fixed before evaluation. This support-only normalization makes cross-category score scales more comparable; it does not establish equality or exchangeability of their distributions. In particular, the $k = 1$ scale is degenerate before clipping, so those cells are interpreted as stress tests rather than ordinary LOIO calibration.

The normalized source score views are then selected without inspecting target labels or target scores. Matched-condition mode selects the source view with the declared target condition and therefore requires condition metadata. Clean-source mode always selects clean source normals. Condition-agnostic mode takes, for each base source image, the median normalized score over all available condition views so that repeated corruptions are not counted as independent observations. Deterministic mismatched-condition routing selects a different condition solely as a negative control. Thus normalization precedes source-view selection or aggregation in the implemented protocol. Only the resulting scalar score archive and category identities are needed downstream; source patch features do not enter certification.

For a fixed target and support seed, the eligible source categories are split before candidate construction into disjoint reference, proposal, and certification sets $\mathcal{R}$, $\mathcal{P}$, and $\mathcal{C}$. The primary allocation is 0.50/0.25/0.25, with deterministic rounding that keeps every partition nonempty. The split is generated from the target-category identity and seed, is held fixed across conditions and k , and never depends on observed scores, target outcomes, or certification losses. All score views from one category remain in the same partition.

3.6. Nested CRESS Threshold Selection

Let $\mathcal{Z}_{\mathcal{R}}$ be the pooled normalized scores from the selected reference-category views. The reference categories define the source-reference rank map

$$p_{\mathcal{R}}(x) = \frac{1 + \sum_{r \in \mathcal{Z}_{\mathcal{R}}} \mathbf{1}\{r \geq \tilde{s}_c(x)\}}{1 + |\mathcal{Z}_{\mathcal{R}}|}. \quad (6)$$

We use the term *source-reference map* deliberately: Eq. (6) is an empirical rank transformation on the source archive, and cross-category conformal validity is not presumed. Its finer grid arises from the larger reference-score pool rather than from the k target supports.

Applying the reference map to the proposal categories produces the proposal p-values $\{p_{\mathcal{R}}(x) : x \in \mathcal{P}\}$. Their distinct values generate an ordered set $\mathcal{T}$ of at most M positive candidate thresholds. If more than M distinct values occur, a deterministic evenly spaced subset of their ordered indices is retained. Certification data never influence this candidate set. The same reference map is then applied to every certification image. Given a candidate $\tau \in \mathcal{T}$, its image-unit loss is

$$L_x(\tau) = \mathbf{1}\{p_{\mathcal{R}}(x) \leq \tau\}. \quad (7)$$

All certification images are declared normal, so an alarm is a false alarm. For a certification category c , the category-unit loss is

$$L_c(\tau) = \frac{1}{n_c} \sum_{x \in c} L_x(\tau) \in [0, 1]. \quad (8)$$

The image unit targets a fixed source-image archive mixture; the category unit targets the mean loss for a new draw from a declared source-category meta-population. They are different estimands and are not interchangeable.

Let A be the number of reported nominal levels and let $n = |\mathcal{C}|$ be the number of certification categories. For each candidate threshold, define the empirical mean category loss

$$\bar{L}_c(\tau) = \frac{1}{n} \sum_{c \in \mathcal{C}} L_c(\tau). \quad (9)$$

The category analysis then uses the family-adjusted Hoeffding category-risk upper confidence bound (UCB)

$$U_H(\tau) = \min \left\{ 1, \bar{L}_c(\tau) + \sqrt{\frac{\log(2AM/\delta)}{2n}} \right\}. \quad (10)$$

For independent Bernoulli image alarms, the fixed-archive sensitivity uses the exact one-sided Clopper–Pearson upper bound with tail probability $\delta/(2AM)$. The factor $2A$ allocates the family error across both unit definitions and all reported levels; M allocates it across the finite candidate family. In the selection rule below, U denotes U_H for category certification and the corresponding Clopper–Pearson bound for the image-unit sensitivity. CRESS returns

$$\tau^* = \max(\{\tau \in \mathcal{T} : U(\tau) \leq \alpha\} \cup \{0\}). \quad (11)$$

The value $\tau^* = 0$ is a deterministic fail-closed fallback because $p_R(x) > 0$. After threshold selection, a target image can be operated on by

$$a_{\tau^*}(x) = \mathbf{1}\{p_R(x) \leq \tau^*\}, \quad (12)$$

where the target score is normalized only by the target support statistics. Target test images and labels enter none of the reference, proposal, certification, or threshold-selection stages.

3.7. Certificate Scope, Feasibility, and Transfer Boundary

Proposition 2 (conditional source certificate). Conditional on the reference and proposal stages, suppose certification units are independent draws from the declared source-unit population: Bernoulli losses for the image analysis, or bounded losses in $[0, 1]$ for the category analysis. Then, with probability at least $1 - \delta$, simultaneously across the A levels and both unit definitions, the corresponding source risk of every selected threshold is at most its α .

Proof. For a fixed level, unit, and candidate, the one-sided Clopper–Pearson construction or Hoeffding’s inequality fails with probability at most $\delta/(2AM)$. A union bound over M candidates, A levels, and two units gives total failure probability at most δ ; independence between these bounds is unnecessary. Selection occurs only on this simultaneous event, and the zero fallback has zero alarm risk. $\square$

Proposition 3 (Hoeffding feasibility). Under the category-level Hoeffding rule, a positive candidate can pass at level α only if

$$n \geq \frac{\log(2AM/\delta)}{2\alpha^2}. \quad (13)$$

Proof. Because $\bar{L}_C(\tau) \geq 0$, the unclipped upper bound is at least $\sqrt{\log(2AM/\delta)/(2n)}$. Requiring it not to exceed α and rearranging proves the claim. $\square$

Proposition 4 (distribution-free all-zero lower bound). Let $\beta \in (0, 1)$ and let $L_1, \dots, L_n$ be iid losses from an arbitrary distribution P on $[0, 1]$, with mean μ_P . Let $U_n : [0, 1]^n \rightarrow [0, 1]$ be any deterministic upper confidence bound satisfying

$$\inf_P \Pr_{P^n} \{\mu_P \leq U_n(L_1, \dots, L_n)\} \geq 1 - \beta.$$

Then

$$U_n(0, \dots, 0) \geq 1 - \beta^{1/n}. \quad (14)$$

Table 1

Best-case minimum independent certification categories at $\delta = 0.05$, assuming zero observed category loss. “DF floor” is the distribution-free lower bound in Proposition 4: first without multiplicity ($\beta = \delta$), then under the frozen allocation $\beta = \delta/(2AM)$ with $A = 3$. “Hoeffding” is Proposition 3.

rule / candidates	$\alpha = .20$	$\alpha = .10$	$\alpha = .05$
DF floor, no multiplicity	14	29	59
DF floor, $M = 1$	22	46	94
DF floor, $M = 5$	29	61	125
DF floor, $M = 20$	35	74	152
Hoeffding, $M = 1$	60	240	958
Hoeffding, $M = 5$	80	320	1,280
Hoeffding, $M = 20$	98	390	1,557

Consequently, even after observing zero loss in every certification category, $U_n \leq \alpha$ is possible only if

$$n \geq \frac{\log \beta}{\log(1 - \alpha)}.$$

Proof. Write $u = U_n(0, \dots, 0)$ and suppose, for contradiction, that $u < 1 - \beta^{1/n}$. Choose a Bernoulli distribution with mean q such that $u < q < 1 - \beta^{1/n}$. Its all-zero sample has probability $(1 - q)^n > \beta$. On that event $U_n = u < q = \mu_P$, so the bound fails with probability greater than β , contradicting uniform $(1 - \beta)$ coverage. Rearranging $1 - \beta^{1/n} \leq \alpha$ gives the sample-size condition. $\square$

Proposition 4 shows that the frozen failure is not solely an artifact of Hoeffding looseness. Even for one fixed candidate, one level, no image-unit co-reporting, and no multiplicity penalty ($\beta = \delta = 0.05$), any deterministic uniformly valid distribution-free bound based only on iid category losses needs at least 14, 29, and 59 categories at $\alpha = 0.20, 0.10$, and 0.05 . Under the frozen allocation $\beta = \delta/(2AM)$ with $A = 3$ and the most favorable $M = 1$, the corresponding lower limits are 22, 46, and 94; the declared Hoeffding rule requires 60, 240, and 958.

Table 1 is a design calculation rather than an empirical result. Proposition 4 is limited to deterministic, uniformly valid, distribution-free upper bounds based solely on iid bounded category losses. Parametric or hierarchical assumptions, informative side data, and randomized confidence procedures require separate validity arguments. Adding source images or repeated support seeds may refine the fixed-archive image analysis, but cannot increase the number of independent certification categories.

Corollary 1 (conditional target transfer). On the event in Proposition 2, target risk is also at most α if, for every proposed threshold, the target normal alarm probability is no larger than the corresponding source-unit risk. The same conclusion follows if the target category is exchangeable with the certification-category meta-population.

The dominance or category-exchangeability condition is an assumption, not a consequence of normalization, source-view selection, or the rank transformation in Eq. (6). Thus

τ^* carries a source-domain certificate. It becomes a target-category guarantee only under the additional transfer condition. Direct pooled-source conformal, which uses the complete selected source pool with $\tau = \alpha$ and no disjoint proposal or certification stage, is retained as an uncertified empirical reference.

4. Experiments

4.1. Datasets and Few-Shot Splits

We conduct evaluations on MVTec AD [37] and VisA [38]; the strict external-transfer audit additionally incorporates MPDD. MVTec AD comprises industrial object and texture categories with pixel-level defect masks. VisA contains a broader range of industrial categories and serves as the primary benchmark for full-scale corruption and calibration-shift. MPDD provides six metal-part categories for an external MVTec-to-MPDD stress test; within-MPDD category certification is not pursued because fewer than six non-target categories remain for a three-way source split. For each category, support sets are sampled from the normal training partition using fixed seeds. The CRESS source archive is separate: it consists of held-out known-normal evaluation images from non-target source categories (the test/good partition for MVTec and the corresponding normal partition for VisA and MPDD). Source normal status is therefore used, but target test labels never enter reference construction, threshold proposal, or certification. Unless otherwise specified, $k \in \{1, 2, 4, 8\}$ is employed for the clean accuracy-storage and strict nested studies, whereas the earlier corruption tables use $k \in \{4, 8\}$. The $k = 1$ strict cells use the declared patch-split support calibration fallback and are interpreted as stress tests; LOIO calibration itself requires $k \geq 2$.

4.2. Evaluation Protocols

We structure the experiments into six protocol groups. The first five address clean accuracy and storage, PCA64 and PCA128 sensitivity, corruption reliability, secondary reliability-view comparisons, and attainable operating levels. The sixth is the frozen strict nested protocol, which uses $k \in \{1, 2, 4, 8\}$, seeds from 0 to 4, $\alpha \in \{0.05, 0.10, 0.20\}$, and clean, Gaussian-noise, blur, brightness/contrast, and JPEG conditions. Each cell contains at most 120 evaluation images, with $\rho = 0.01$, PCA64, $\delta = 0.05$, and up to 20 candidate thresholds. Raw scores are first normalized using support-only statistics; condition-specific source views are then selected or aggregated as defined in Section 3.5. For every target category and seed, the eligible source categories are deterministically split using the frozen 0.50/0.25/0.25 reference/proposal/certification allocation. This gives 7/3/4 categories for a 14-category within-MVTec source pool, 5/3/3 for an 11-category within-VisA pool, and 7/4/4 for the 15-category MVTec transfer pool. The same category partition is reused across conditions and k . The protocol evaluates MVTec and VisA within-dataset, MVTec-to-VisA, and MVTec-to-MPDD under matched-condition, clean-source,

condition-agnostic, and mismatched negative-control source selection. The operational gate requires a nonzero category-certified threshold in at least 80% of target cells, a target false-alarm rate (FAR) of at most $\alpha + 0.02$, nonzero power, at least 80% no-harm, and a positive lower bound on simultaneous power gain. Zero thresholds and failed cells are retained.

4.3. Baselines and Ablations

We compare against a controlled nearest-neighbor memory bank using DINOv2, a SubspaceAD-style PCA residual baseline, LOIO conformal, weighted conformal, and the post-hoc calibrators described below. For the source-assisted study, direct pooled-source conformal is an uncertified baseline: it uses the complete selected source pool with $\tau = \alpha$, but performs no reference, proposal, or certification split and applies no UCB gate. The controlled memory-bank implementation is not an official reproduction of Patch-Core or AnomalyDINO; it maintains the frozen DINOv2 feature pipeline and common split to isolate scoring and storage trade-offs. Official AnomalyDINO code is also run on MVTec and VisA under its released protocol and reported as a separate accuracy reference. WinCLIP [12] is included only through published values because no official implementation is available. Official SubspaceAD representative runs serve as a novelty guardrail: the frozen DINOv2 subspace residual is inherited, and the contribution begins with the reliability analysis downstream of that score.

4.4. Secondary Calibration and Routing Baselines

The post-hoc calibration families are secondary baselines rather than components of CRESS. Scalar Platt maps $s_c(x)$ directly to a probability. Vector Platt uses

$$\phi(x) = \left[s_{\text{pca}}(x), s_{\text{head}}(x), \left| \tilde{s}_{\text{pca}}(x) - \tilde{s}_{\text{head}}(x) \right| \right], \quad (15)$$

where s_{head} is the maximum patch probability from a small synthetic-anomaly MLP ($384 \rightarrow 256 \rightarrow 1$, dropout 0.1). The head is trained only on support normal patches and synthetic perturbed patches and never changes the ranking score. Shift-Aware Platt appends five pre-specified shift descriptors, while weighted Platt uses density-ratio weights from a logistic domain classifier. Temperature scaling, isotonic regression, histogram binning, and scalar Platt are fit on the same label-free normal/synthetic calibration set. The real-anomaly validation mode is retained only as an explicitly labeled upper bound.

Exploratory reliability-view selection compares fixed LOIO with validation-ECE and simple no-label rules based on shift descriptors or effective sample size. These analyses are diagnostic: they do not define CRESS, do not alter $s_c(x)$, and do not establish safety under distribution shift. Entropy-ranked abstention, expert disagreement, conformal confidence, effective sample size, and FGSM sensitivity are likewise reported only as secondary deployment or fragility diagnostics.

4.5. Metrics

Ranking metrics are distinguished from reliability metrics. AUROC and AP are calculated from the raw PCA

or subspace residual score $s(x)$. Pixel-level localization is assessed using pixel AUROC and AU-PRO when masks are available. Reliability is evaluated operationally by empirical false-alarm rate at fixed α , detection power, alarm precision, discrete uniformity of normal p-values, and risk-versus-coverage curves. Expected calibration error (ECE), Brier score, and negative log-likelihood (NLL) are reported as secondary metrics based on the selected reliability probability or probability-like conformal view. Storage is quantified as retained reference or model state per class. Robustness experiments include corruption shift and a label-aware PCA-residual FGSM surrogate, with the latter serving exclusively as a fragility diagnostic.

4.6. Reproducibility and Claim Rules

All experiments were conducted on a single NVIDIA RTX 5090 GPU using PyTorch 2.12.1, with DINOv2 patch features cached in fp32 format. For reproducibility, the frozen commit, DINOv2 source and weight hashes, environment specifications, dataset counts, per-image predictions, base-image identities, support manifests, corruption parameters, source partitions, candidate-threshold bounds, and SHA-256 checksums are documented. Automated integrity checks encompass 1,500 MVTEC, 1,200 VisA, and 600 MPDD view cells, identifying duplicate cells, missing support statistics, or overlap between support and test sets. Reliability methods cannot claim ranking improvement unless the raw score is altered; image-level and category-level certificates are not interchangeable.

5. Results

5.1. Ranking Substrate and Storage

Table 2 presents clean image-level ranking performance and category-specific storage. The official AnomalyDINO reproduction achieves the highest image AUROC at each reported k on both datasets, but follows its released evaluation protocol and therefore serves as an external accuracy reference rather than evidence of superiority under our controlled protocol. WinCLIP is likewise a cited reference: its paper reports AUROC and AUPR at one and four shots, but not an AP implementation verified to match ours. We therefore mark its AP entries N/C instead of mixing the reported AUPR values into the controlled AP comparison. WinCLIP also does not report ECE under our calibration protocol, category-specific storage under our accounting, or an eight-shot result.

Compared with the controlled DINOv2 nearest-neighbor ranker, PCA64 yields lower AUROC by 0.010, 0.005, and 0.003 on MVTEC and higher AUROC by 0.019, 0.008, and 0.007 on VisA at $k = 1, 4, 8$, respectively, while requiring 0.472 MiB of category-specific state instead of 2.005 to 6.000 MiB. PCA64 is therefore a compact supporting ranker, not a ranking contribution or state-of-the-art result. On VisA, PCA128 increases class-averaged AUROC over PCA64 by 0.011, 0.015, and 0.017, with storage rising to 0.566 MiB. Boldface is restricted to methods evaluated

under our common protocol; external AnomalyDINO and WinCLIP references are excluded.

The controlled nearest-neighbor ECE values diagnose a failed scalar calibration fit rather than a ranking failure. Exact support-patch self-matches at low k and the 4096-patch subsampling regime at larger k coincide with near-constant test probabilities. These degenerate probabilities are not used as a competitive calibration baseline.

Storage and compute accounting. Storage excludes the shared frozen DINOv2 ViT-S/14 backbone, which is stored once for all categories. The 0.472 MiB PCA64-based reliability state comprises the PCA mean and 64-component basis, the synthetic-anomaly MLP head, and feature-normalization statistics. Expanding to 128 components adds 0.094 MiB. The controlled memory bank stores up to 4096 fp32 vectors of dimension 384, or 6.000 MiB. With cached features, PCA-residual scoring takes approximately 1.3 ms per image, compared with 5 to 13 ms for controlled nearest-neighbor scoring. These timings exclude shared backbone inference. CRESS subsequently retains scalar normalized reference scores and partition metadata rather than source patch features.

The clean benchmark uses max pooling, whereas the conformal pipeline uses the top-1% mean. Table 16 in Appendix A shows that both choices lie within 0.01 to 0.02 AUROC of the best tested setting. Thus pooling does not drive the reported ranking comparison, although invariance outside the tested grid is not claimed.

5.2. Target-Only Resolution and False-Alarm Behavior

Table 3 isolates the structural resolution limit. With $k = 4$, the smallest p-value is $1/(k+1) = 0.2$, so nominal levels 0.01, 0.05, and 0.10 cannot raise an alarm. With $k = 8$, the floor decreases to $1/9 \approx 0.111$, but the same three levels remain below it. Their zero FAR and detection values are therefore finite-resolution artifacts, not evidence of effective conservative control.

At nominal $\alpha = 0.20$, the effective threshold is 0.20 for $k = 4$ but only $1/9$ for $k = 8$, because $1/9$ is the largest $k = 8$ grid point not exceeding 0.20. On VisA, the aggregate FAR/detection pairs are 0.152/0.599 at $k = 4$ and 0.111/0.537 at $k = 8$; on MVTEC they are 0.373/0.886 and 0.261/0.907. These values first pool images over categories and seeds within each corruption and then average the four corruption-specific rates. The same nominal α can therefore induce different effective thresholds across k , while the observed FAR itself is not restricted to the p-value grid.

Table 4 resolves this aggregate by corruption. VisA remains below the nominal level in all cells, with FAR from 0.142 to 0.157 at $k = 4$ and from 0.103 to 0.123 at $k = 8$. MVTEC is less stable: at $k = 4$, FAR ranges from 0.306 to 0.463; at $k = 8$, blur and brightness/contrast approach α at 0.194 and 0.188, while Gaussian noise and JPEG remain anti-conservative at 0.355 and 0.307. Increasing k mitigates but does not eliminate the mismatch. Precision is descriptive because it depends on benchmark anomaly prevalence.

Table 2

Clean image-level ranking performance and category-specific storage. Controlled DINOv2 NN and PCA64/PCA128 share our feature, split, metric, and storage-accounting protocol and are averaged over classes and five seeds. The official AnomalyDINO reproduction uses three seeds under its released protocol; WinCLIP values are reported in [12]. Bold marks the best value only among common-protocol rows in each dataset/ k block. “–” denotes not evaluated, N/C not directly comparable, and N/R not reported.

Dataset	k	Method	AUROC $\uparrow$	AP $\uparrow$	ECE $\downarrow$	MiB $\downarrow$
MVTec	1	Official AnomalyDINO	0.965	0.982	–	–
		WinCLIP (reported)	0.931	N/C	N/R	N/R
		Controlled DINOv2 NN	0.914	0.952	0.277	2.005
		PCA64 (ours)	0.904	0.951	0.274	0.472
	4	Official AnomalyDINO	0.976	0.984	–	–
		WinCLIP (reported)	0.952	N/C	N/R	N/R
		Controlled DINOv2 NN	0.942	0.967	0.693	6.000
		PCA64 (ours)	0.937	0.967	0.214	0.472
	8	Official AnomalyDINO	0.980	0.990	–	–
		Controlled DINOv2 NN	0.948	0.970	0.693	6.000
		PCA64 (ours)	0.945	0.972	0.154	0.472
VisA	1	Official AnomalyDINO	0.857	0.866	–	–
		WinCLIP (reported)	0.838	N/C	N/R	N/R
		Controlled DINOv2 NN	0.804	0.809	0.434	2.005
		PCA64 (ours)	0.823	0.834	0.429	0.472
		PCA128 (ours)	0.834	0.842	0.430	0.566
	4	Official AnomalyDINO	0.912	0.918	–	–
		WinCLIP (reported)	0.873	N/C	N/R	N/R
		Controlled DINOv2 NN	0.862	0.861	0.543	6.000
		PCA64 (ours)	0.870	0.883	0.284	0.472
		PCA128 (ours)	0.885	0.895	0.278	0.566
	8	Official AnomalyDINO	0.926	0.929	–	–
		Controlled DINOv2 NN	0.873	0.870	0.533	6.000
		PCA64 (ours)	0.880	0.891	0.205	0.472
		PCA128 (ours)	0.897	0.905	0.170	0.566

Table 3

Target-only LOIO operating points. The floor is the smallest p-value, $1/(k+1)$; “attainable” is the largest p-value grid point not exceeding the nominal α , or zero if none exists. Within each corruption, FAR and detection are pooled over classes and seeds; the table then averages the four corruption-specific rates.

dataset	k	floor	nominal α	attainable	FAR	detection
VisA (full)	4	0.200	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.200	0.152	0.599
	8	0.111	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.111	0.111	0.537
	4	0.200	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.200	0.373	0.886
MVTec (full)	8	0.111	0.01	0.000	0.000	0.000
			0.05	0.000	0.000	0.000
			0.10	0.000	0.000	0.000
			0.20	0.111	0.261	0.907

An fp32 representation of $1/5$ may equal 0.20000000298, so exact comparison with 0.2 can inadvertently exclude floor-level alarms. All reported thresholding applies a fixed numerical tolerance.

5.3. Normal-P-Value Shift Audit

Conformal p-values satisfy $P(p \leq \alpha) \leq \alpha$ for normal data only under the relevant exchangeability conditions.

Table 4

Target-only LOIO false-alarm rate (FAR), anomaly-detection rate, and precision at the first reported attainable level, $\alpha=0.20$. Results are pooled over classes and seeds and reported separately by dataset, corruption, and k . Alarms fire at $p \leq \alpha$ using the comparison tolerance described in the text.

dataset	corruption	k	FAR	detection	precision
VisA (full)	blur	4	0.155	0.610	0.806
		8	0.109	0.546	0.841
	bright/contr.	4	0.155	0.595	0.803
		8	0.110	0.533	0.836
	Gauss. noise	4	0.142	0.585	0.813
		8	0.103	0.522	0.843
	JPEG	4	0.157	0.604	0.803
		8	0.123	0.547	0.825
	blur	4	0.306	0.887	0.872
		8	0.194	0.908	0.916
MVTec (full)	bright/contr.	4	0.313	0.887	0.869
		8	0.188	0.908	0.919
	Gauss. noise	4	0.463	0.879	0.816
		8	0.355	0.898	0.855
	JPEG	4	0.410	0.890	0.836
		8	0.307	0.914	0.875

Because a k -score p -value lies on the grid $\{j/(k+1)\}$, a continuous Kolmogorov–Smirnov test against $U(0,1)$ is inappropriate. Figure 2 instead compares the empirical normal- p -value CDF with the ideal discrete-uniform reference at each attainable point. VisA at $k=4$ is conservative, with the largest negative gap of -0.10 at 0.4 , whereas VisA at $k=8$ and MVTec at both k values become anti-conservative under corruption. For example, MVTec $k=4$ under Gaussian noise has $\hat{F}(0.2) = 0.46$ compared with the reference value 0.20 .

Rows share calibration sets, categories, and repeated seeds, so the audit uses a cluster-preserving exchangeable Monte Carlo diagnostic rather than an iid confirmatory test. The results indicate that corrupted normal residuals increase relative to clean support calibration scores, but do not identify a causal mechanism. A matched-LOIO audit on representative MVTec $k=4$ categories increases clean FAR from 0.17 to 0.25 while leaving power essentially unchanged. Fold asymmetry therefore does not account for the observed anti-conservative behavior.

5.4. Strict CRESS Category Certification

Table 5 summarizes the frozen strict nested audit. The primary category-level result is uniformly negative. The 960 gate configurations comprise four source-to-target jobs, four source-view modes, four k values, three α levels, and five conditions. None reaches the required 80% nonzero-threshold rate: every category-certified target cell receives $\tau^* = 0$. Only four certification categories are available per MVTec and transfer cell and three per VisA cell. The smallest candidate UCB is 0.950 on MVTec, 1.000 on VisA, 0.961 for MVTec-to-VisA, and 0.986 for MVTec-to-MPDD, all exceeding the largest tested $\alpha = 0.20$.

This outcome is not a GPU or optimization failure. Proposition 3 shows that with $A = 3$, $\delta = 0.05$, and even $M = 1$, a zero-loss Hoeffding gate needs at least 60 categories at $\alpha = 0.20$. Proposition 4 shows that the finding is not solely due to Hoeffding looseness: without multiplicity, any deterministic uniformly valid distribution-free 95% upper bound based only on iid category losses needs at least 14 all-zero categories to reach 0.20 ; the frozen family allocation increases this requirement to 22. Both exceed the available three or four categories.

The image-unit block changes the estimand to a fixed source archive. With 72 to 191 image units per cell, the smallest UCB decreases to 0.039 to 0.048 . Positive thresholds occur in 36.7% of MVTec, 60.4% of VisA, 41.5% of MVTec-to-VisA, and 37.0% of MVTec-to-MPDD cells. These rates do not meet the 80% operational criterion, and the additional images do not create independent categories. The image result is therefore a fixed-archive sensitivity analysis, not a new-category certificate.

Table 6 reports direct pooled-source conformal as an un-certified reference. It uses the complete selected source pool with $\tau = \alpha$ and no disjoint reference, proposal, or certification categories. Under matched-condition source selection, it has nonzero power in all four source-to-target jobs, but mean FAR exceeds nominal within both MVTec and VisA. Transfer is more conservative but also less powerful. Direct pooling therefore supplies finer operating points than target-only LOIO, while characterizing empirical risk under the selected source mixture rather than certified new-category risk.

5.5. Secondary Calibration Diagnostics

Table 7 reports the full VisA calibration summaries under a fixed ranking score. Across 56,000 corruption-view

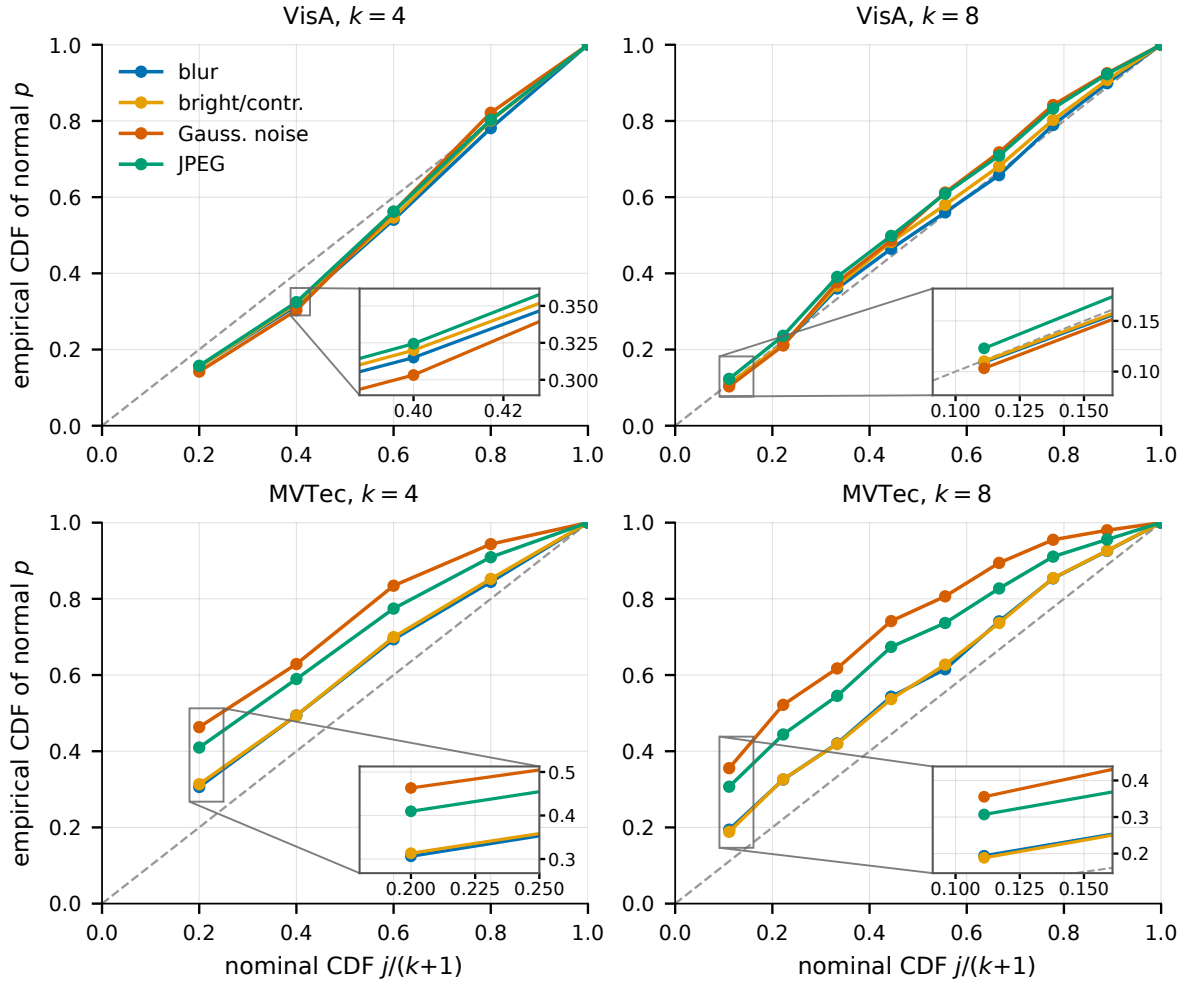


Figure 2: Empirical CDF of normal-image LOIO p-values at each attainable point $j/(k+1)$ versus the idealized exchangeable reference (dashed diagonal), by corruption. Points above the diagonal indicate excess false alarms relative to the reference; lower-right insets enlarge the first informative grid region.

Table 5

Frozen strict nested CRESS audit over four source-view modes, $k \in \{1, 2, 4, 8\}$, three α levels, five seeds, and five conditions. “Min UCB” is the smallest observed candidate upper bound. Nonzero is the fraction of target cells receiving $\tau > 0$. Category and image rows target different estimands.

source $\rightarrow$ target	certification unit	units/cell	min UCB	nonzero	interpretation
MVTec $\rightarrow$ MVTec	category	4	0.950	0.000	new-category gate fails
VisA $\rightarrow$ VisA	category	3	1.000	0.000	new-category gate fails
MVTec $\rightarrow$ VisA	category	4	0.961	0.000	new-category gate fails
MVTec $\rightarrow$ MPDD	category	4	0.986	0.000	new-category gate fails
MVTec $\rightarrow$ MVTec	fixed-archive image	75 to 191	0.039	0.367	sensitivity only
VisA $\rightarrow$ VisA	fixed-archive image	150 to 180	0.042	0.604	sensitivity only
MVTec $\rightarrow$ VisA	fixed-archive image	72 to 159	0.048	0.415	sensitivity only
MVTec $\rightarrow$ MPDD	fixed-archive image	81 to 190	0.040	0.370	sensitivity only

rows, LOIO has pooled ECE 0.0766, with 0.0391 at $k = 4$ and 0.1140 at $k = 8$. Weighted conformal is substantially worse at $k = 4$ and approaches LOIO at $k = 8$, but has higher Brier score and NLL. AUROC and AP remain identical because neither reliability transformation changes $s_c(x)$.

On the corresponding MVTec study, LOIO has overall ECE 0.0684 while maintaining image AUROC 0.912. LOIO is lower than Vector and Shift-Aware Platt in all eight displayed k -by-corruption cells, but this is a descriptive pattern rather than a multiplicity-adjusted confirmatory result. ECE remains secondary and does not imply false-alarm control.

Table 6

Direct pooled-source conformal under matched-condition source selection. FAR and power are means over target class, $k \in \{1, 2, 4, 8\}$, seeds from 0 to 4, and five conditions. The baseline uses the complete selected source pool with $\tau = \alpha$ and has no reference, proposal, or certification split.

source $\rightarrow$ target	α	FAR	power
MVTec $\rightarrow$ MVTec	0.05	0.075	0.404
	0.10	0.131	0.566
	0.20	0.242	0.754
VisA $\rightarrow$ VisA	0.05	0.060	0.320
	0.10	0.111	0.466
	0.20	0.208	0.639
MVTec $\rightarrow$ VisA	0.05	0.028	0.171
	0.10	0.048	0.282
	0.20	0.099	0.441
MVTec $\rightarrow$ MPDD	0.05	0.030	0.186
	0.10	0.066	0.283
	0.20	0.157	0.423

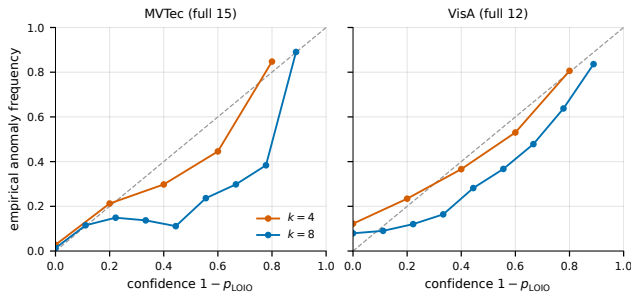


Figure 3: LOIO reliability diagrams on the full MVTec (left) and VisA (right) corruption benchmarks, pooled over categories, seeds, and corruptions. Points occupy the $k + 1$ attainable confidence levels; positions below the diagonal indicate overconfidence.

Figure 3 shows distributional structure hidden by scalar ECE. The attainable confidence values are sparse, and the pooled $k = 8$ curve lies farther below the diagonal in the upper-confidence region than the $k = 4$ curve. These aggregates do not imply that every category shares the same deviation.

Table 8 in Appendix A and Figure 4 give the cellwise breakdown. LOIO is lowest in all four VisA $k = 4$ cells and under Gaussian noise at $k = 8$; weighted conformal is lower for the other three VisA $k = 8$ corruptions. LOIO is lowest in every displayed MVTec cell. This cellwise summary is more accurate than claiming uniform superiority.

Table 9 compares the secondary label-free calibrators using per-cell ECE. LOIO has lower mean ECE in most comparisons, although the margin narrows at $k = 8$. Historical unadjusted significance values are omitted because a complete multiplicity-adjusted analysis is unavailable. ECE on $1 - p$ also changes with anomaly prevalence: on VisA it decreases from 0.404 at 1% prevalence to 0.149 at 50%, and the ordering between LOIO and weighted methods reverses.

Operational FAR, power, and precision therefore carry the primary interpretation.

5.6. Historical, Non-Independent CRESS Sensitivity

The historical analyses reuse source evidence and employ pointwise rather than simultaneous intervals. They are retained in the Appendix as empirical sensitivity checks and do not supersede the strict category result.

On MVTec $k = 4$, historical matched-condition CRESS has pooled power 0.216 and 0.416 at $\alpha = 0.05$ and 0.10, with FAR 0.050 and 0.105. At $k = 8$, pooled power is 0.547 and 0.693, while pooled FAR is 0.052 and 0.121. Condition-level exceptions, particularly Gaussian noise and JPEG, prevent a uniform control claim. The paired changes in FAR and power in Tables 10 and 11 indicate a threshold-selection trade-off rather than superiority over the target-only anchor.

Smoothed randomized p-values and historical CRESS each enable sub-floor alarms, but neither dominates across datasets and operating levels (Table 12). Within-VisA and MVTec-to-VisA source pools likewise display different FAR and power trade-offs (Table 13). Because threshold selection and source assessment are not independent in these analyses, none is interpreted as certified evidence.

5.7. Secondary Deployment and Detector Diagnostics

Entropy-ranked abstention in Figure 5 reduces LOIO ECE from 0.068 to 0.030 on MVTec when the most uncertain 20% of images are withheld, but only from 0.077 to 0.075 on VisA. This dataset-dependent behavior is a selective-operation diagnostic, not a certificate.

Table 14 reports a protocol-locked diagnostic comparison of fixed LOIO and Vector Platt. LOIO lowers ECE in the displayed transfer, leave-one-category-out, and within-split cells, but the table covers representative protocols and ECE only; it does not demonstrate superiority of a learned or weighted route across datasets, shifts, or operational metrics.

Table 15 distinguishes localization from robustness. PCA64 remains competitive with the controlled memory bank in pixel AUROC and AU-PRO, but the small metric-dependent differences do not support a general localization-superiority claim. Under the label-aware PCA-residual FGSM surrogate, image AUROC falls sharply and non-monotonically. These results diagnose sensitivity to the surrogate objective and attack configuration; they do not support adversarial robustness.

6. Limitations

The evidence is limited to MVTec, VisA, and MPDD. The frozen DINOv2 PCA residual is an inherited ranking substrate, while official AnomalyDINO and reported WinCLIP results use different protocols and serve only as external references. The controlled shift study covers four synthetic corruptions, not the full range of temporal drift, sensor replacement, process changes, correlated failures, or deployment prevalence. Moreover, scoring latency is

Table 7

Full VisA conformal reliability under four corruptions, pooled over classes and seeds. The PCA/subspace ranking score is fixed, so AUROC and AP do not change between reliability routes; N is the number of evaluated corruption-view rows. Bold marks the lower ECE, Brier score, and NLL within each group.

Group	Reliability route	N	AUROC	AP	ECE ↓	Brier ↓	NLL ↓
All	LOIO conformal	56000	0.823	0.846	0.077	0.187	0.686
	Weighted conformal	56000	0.823	0.846	0.166	0.244	0.813
$k=4$	LOIO conformal	28000	0.820	0.842	0.039	0.187	0.760
	Weighted conformal	28000	0.820	0.842	0.216	0.259	0.925
$k=8$	LOIO conformal	28000	0.828	0.851	0.114	0.188	0.612
	Weighted conformal	28000	0.828	0.851	0.116	0.229	0.701

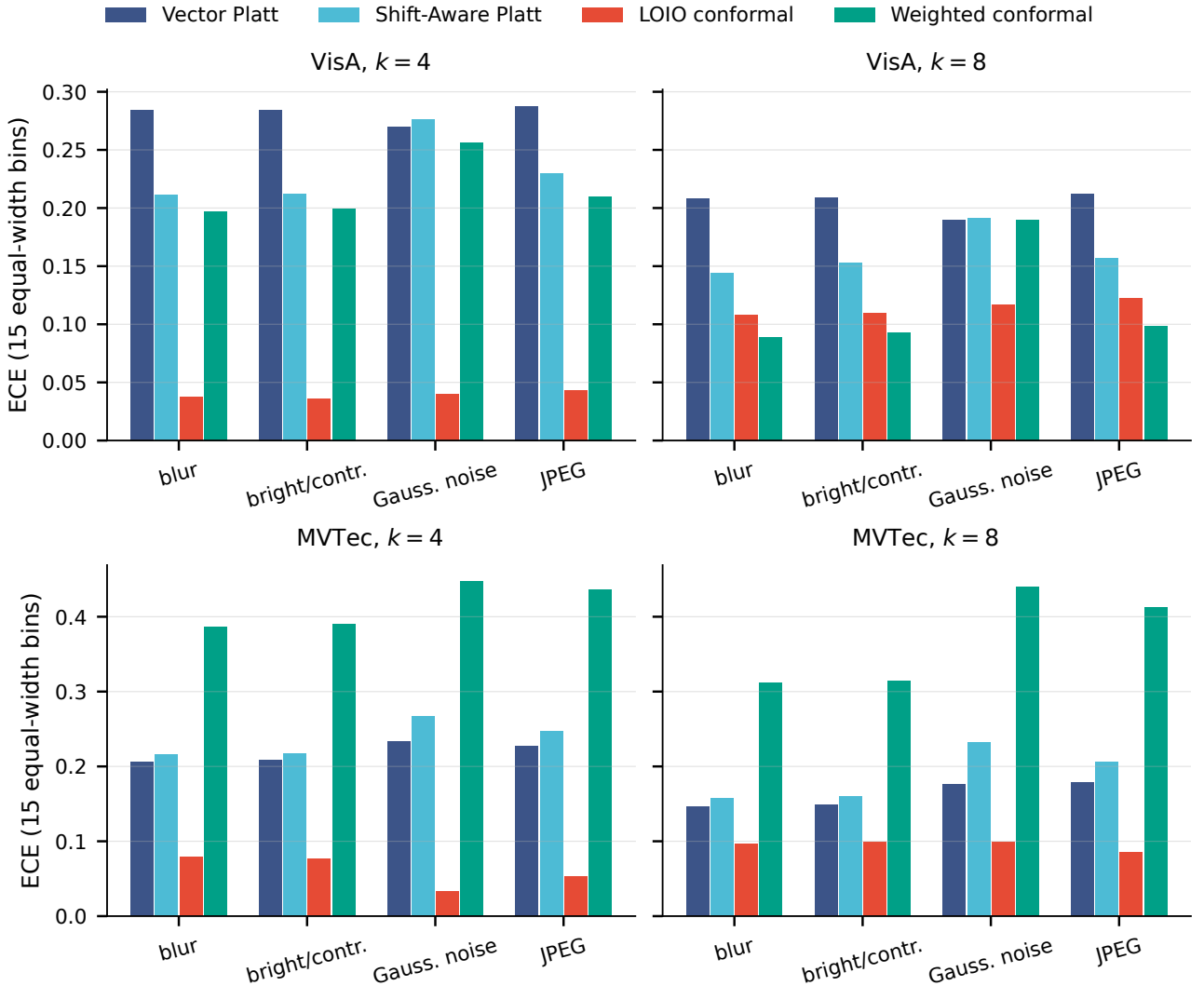


Figure 4: ECE by corruption and reliability route on full VisA (top) and full MVTEC (bottom) at $k = 4$ and $k = 8$. Lower values indicate better calibration under the reported ECE definition.

measured on cached DINOv2 features and excludes backbone inference, preprocessing, and I/O. Generalization to prospective factory operation therefore requires evaluation under real shifts and end-to-end deployment constraints.

Target-only conformal validity requires representative normal supports and the relevant exchangeability conditions. LOIO calibration and full-support test residuals are asymmetric, and their p-values occupy a finite grid; our clustered discrete-reference analyses diagnose departures

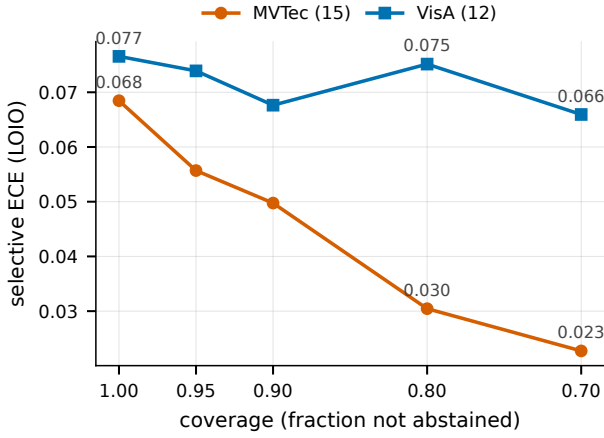


Figure 5: LOIO risk-versus-coverage curves with entropy-ranked abstention on full MVTEC and VisA. Coverage is the retained image fraction; risk is ECE on the retained subset.

from the exchangeable pattern but do not restore validity when exchangeability fails. ECE is additionally sensitive to prevalence and binning, so it remains descriptive rather than a false-alarm guarantee. The FGSM results define a separate boundary: robustness to the evaluated corruptions does not imply adversarial robustness.

The certification claims are narrower still. Historical CRESS results reuse source evidence and are empirical diagnostics, whereas the independent strict evaluation returns $\tau = 0$ for every category-certified target cell because only three or four certification categories are available. The distribution-free all-zero lower bound explains why additional images or seeds cannot repair this category shortage; assumption-dependent parametric or hierarchical alternatives would require their own validity arguments. Fixed-archive image certificates target an archive mixture, not a new-category risk. Finally, matched source selection assumes reliable condition metadata, and source-to-target control still requires source-risk dominance or category exchangeability. Thus the current evidence supports an auditable source-assisted procedure and its feasibility boundary, not a deployable certified alarm for an arbitrary target category.

7. Conclusion and future work

We have presented an auditable separation between low-storage anomaly ranking and few-shot alarm reliability. Three conclusions have emerged. First, a target-only conformal alarm built from k calibration scores cannot fire below $\alpha = 1/(k+1)$; zero alarms below this floor are a resolution artifact, not evidence of false-alarm control. Second, crossing the floor does not guarantee stability: corrupted normal inputs can make empirical false-alarm behavior anti-conservative without any change to the underlying ranker. Third, source-assisted category certification is limited by the number of independent categories, not by the number

of images or support seeds. In the strict CRESS evaluation, every category-certified target cell receives $\tau^* = 0$, and all 960 gate configurations fail across MVTEC, VisA, and two transfer directions. This is not merely Hoeffding looseness: even without multiplicity, a deterministic distribution-free 95% category-risk upper bound requires at least 14 all-zero categories to reach $\alpha = 0.20$, compared with the three or four available. Fixed-archive image certificates and historical CRESS results provide sensitivity evidence for different or non-independent estimands; neither establishes certified target-category control.

The practical contribution is therefore an audit discipline rather than an unconditional deployment guarantee: report attainable rather than nominal operating points, test normal false alarms under declared shifts, specify the certification unit before observing results, and count independent categories rather than repeated views. CRESS remains a source-assisted thresholding and certification procedure, but its category-level certificate is infeasible at the archive sizes studied here. Future work must either collect category-rich source archives or introduce pre-specified parametric, hierarchical, or side-information assumptions with independent validity checks; a nominally tighter distribution-free bound cannot overcome the all-zero category limit with only three or four categories. Prospective evaluation should additionally cover real production shifts, temporal dependence, condition-estimation error, and end-to-end deployment cost.

CRedit authorship contribution statement

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets analyzed during the current study are publicly available: MVTEC AD (<https://www.mvtec.com/company/research/datasets/mvtec-ad>) and VisA (<https://github.com/amazon-science/spot-diff>). The final review package will include derived per-image prediction tables, evaluation summaries, sampling and source-partition manifests, and check-sums; the reviewer-accessible repository identifier must be inserted before submission.

Code availability

The experimental code includes the conformal protocols, calibrators, corruption pipeline, artifact audits, and table/figure generators. A reviewer-accessible repository and immutable commit identifier must be inserted here before submission.

Declaration of generative AI and AI-assisted technologies

During manuscript preparation, the authors used OpenAI Codex for code review, test generation, statistical-claim auditing, and language editing. No experimental observation was generated or altered by the tool. The authors reviewed the resulting code and text and take responsibility for the content.

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A. Supplementary tables

This appendix collects supporting tables summarized in the main text: the per-corruption VisA ECE breakdown and standard-calibrator comparison (Tables 8 and 9); the historical, non-independent CRESS sensitivity studies from Table 10 to Table 13; protocol-routing, localization, and fragility diagnostics (Tables 14 and 15); the aggregation-sensitivity ablation (Table 16); and the novelty-positioning summary (Table 17).

The exploratory reliability-view selection analysis was conceptually inspired by the shared-versus-specialized selection motif in SAGE [39]. CRESS itself does not use a gated-expert architecture: its reference, proposal, and certification partitions serve statistical roles, and its source-view modes are deterministic protocol choices rather than learned expert gates.

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Table 8

ECE by reliability route on the full VisA corruption benchmark, pooled over classes and seeds and reported separately for each combination of k and corruption. Lower is better, and bold marks the lowest ECE in each cell. Figure 4 gives the corresponding visual comparison for VisA and MVTec.

k	Corruption	Vector	Shift-aware	LOIO	Weighted
4	blur	0.284	0.211	0.037	0.197
	brightness/contrast	0.285	0.212	0.036	0.199
	gaussian noise	0.270	0.276	0.040	0.256
	jpeg	0.288	0.230	0.043	0.210
8	blur	0.208	0.144	0.108	0.089
	brightness/contrast	0.209	0.153	0.109	0.093
	gaussian noise	0.190	0.191	0.117	0.189
	jpeg	0.212	0.156	0.123	0.099

Table 9

LOIO conformal reliability versus label-free calibrators on the historical full corruption benchmarks. Entries are per-cell ECE mean $\pm$ standard deviation; Δ is LOIO minus baseline (negative favors LOIO), and “LOIO better” is its per-cell win fraction.

dataset / k	baseline	base ECE	LOIO ECE	Δ	LOIO better
MVTec $k=4$	Vector Platt	0.223 $\pm$ 0.095	0.120$\pm$0.054	-0.104	84%
	Shift-Aware Platt	0.241 $\pm$ 0.104	0.120$\pm$0.054	-0.121	88%
	Temperature scaling	0.240 $\pm$ 0.104	0.120$\pm$0.054	-0.120	87%
	Isotonic regression	0.222 $\pm$ 0.088	0.120$\pm$0.054	-0.103	87%
	Histogram binning	0.229 $\pm$ 0.099	0.120$\pm$0.054	-0.110	84%
	Scalar Platt	0.268 $\pm$ 0.127	0.120$\pm$0.054	-0.148	93%
MVTec $k=8$	Vector Platt	0.167 $\pm$ 0.065	0.135$\pm$0.055	-0.033	66%
	Shift-Aware Platt	0.192 $\pm$ 0.088	0.135$\pm$0.055	-0.057	75%
	Temperature scaling	0.177 $\pm$ 0.080	0.135$\pm$0.055	-0.042	69%
	Isotonic regression	0.157 $\pm$ 0.069	0.135$\pm$0.055	-0.023	55%
	Histogram binning	0.162 $\pm$ 0.080	0.135$\pm$0.055	-0.027	61%
	Scalar Platt	0.256 $\pm$ 0.132	0.135$\pm$0.055	-0.121	91%
VisA $k=4$	Vector Platt	0.286 $\pm$ 0.060	0.131$\pm$0.063	-0.155	97%
	Shift-Aware Platt	0.237 $\pm$ 0.087	0.131$\pm$0.063	-0.106	90%
	Temperature scaling	0.312 $\pm$ 0.085	0.132$\pm$0.059	-0.180	98%
	Isotonic regression	0.239 $\pm$ 0.073	0.132$\pm$0.059	-0.106	90%
	Histogram binning	0.301 $\pm$ 0.087	0.132$\pm$0.059	-0.168	96%
	Scalar Platt	0.304 $\pm$ 0.145	0.132$\pm$0.059	-0.171	90%
VisA $k=8$	Vector Platt	0.199 $\pm$ 0.063	0.151$\pm$0.054	-0.048	81%
	Shift-Aware Platt	0.162 $\pm$ 0.075	0.151$\pm$0.054	-0.011	51%
	Temperature scaling	0.247 $\pm$ 0.089	0.158$\pm$0.051	-0.089	75%
	Isotonic regression	0.194 $\pm$ 0.068	0.158$\pm$0.051	-0.035	62%
	Histogram binning	0.242 $\pm$ 0.097	0.158$\pm$0.051	-0.083	69%
	Scalar Platt	0.418 $\pm$ 0.100	0.158$\pm$0.051	-0.259	98%

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Table 10

Historical matched-condition CRESS versus the target-only LOIO anchor on 15 label-stratified MVTec classes ($k=4$, seeds from 0 to 2). Threshold selection and source assessment are not independent. Bold marks cells with nonzero power and $\text{FAR} \leq \alpha + 0.02$.

corruption	α	CRESS (historical)			Target-only anchor		
		FAR	power	prec.	FAR	power	prec.
clean	0.05	0.050	0.267	0.928	0.000	0.000	–
	0.10	0.100	0.470	0.957	0.000	0.000	–
	0.20	0.209	0.768	0.914	0.267	0.863	0.898
blur	0.05	0.048	0.242	0.915	0.000	0.000	–
	0.10	0.099	0.468	0.950	0.000	0.000	–
	0.20	0.207	0.760	0.904	0.269	0.860	0.895
bright/contr.	0.05	0.050	0.254	0.914	0.000	0.000	–
	0.10	0.101	0.467	0.954	0.000	0.000	–
	0.20	0.205	0.769	0.905	0.268	0.863	0.898
Gauss. noise	0.05	0.048	0.184	0.934	0.000	0.000	–
	0.10	0.105	0.295	0.929	0.000	0.000	–
	0.20	0.226	0.495	0.914	0.460	0.860	0.844
JPEG	0.05	0.051	0.134	0.911	0.000	0.000	–
	0.10	0.121	0.381	0.936	0.000	0.000	–
	0.20	0.235	0.633	0.917	0.434	0.867	0.854

Table 11

Historical matched-condition CRESS versus the target-only LOIO anchor on 15 label-stratified MVTec classes ($k=8$, seeds from 0 to 2). As in the $k=4$ historical analysis, threshold selection and source assessment are not independent. Bold marks cells with nonzero power and $\text{FAR} \leq \alpha + 0.02$.

corruption	α	CRESS (historical)			Target-only anchor		
		FAR	power	prec.	FAR	power	prec.
clean	0.05	0.053	0.764	0.976	0.000	0.000	–
	0.10	0.110	0.833	0.936	0.000	0.000	–
	0.20	0.216	0.898	0.917	0.182	0.903	0.928
blur	0.05	0.050	0.751	0.976	0.000	0.000	–
	0.10	0.110	0.813	0.957	0.000	0.000	–
	0.20	0.221	0.900	0.917	0.199	0.904	0.922
bright/contr.	0.05	0.051	0.770	0.976	0.000	0.000	–
	0.10	0.107	0.831	0.958	0.000	0.000	–
	0.20	0.223	0.895	0.916	0.185	0.904	0.926
Gauss. noise	0.05	0.035	0.155	0.962	0.000	0.000	–
	0.10	0.138	0.398	0.951	0.000	0.000	–
	0.20	0.232	0.749	0.917	0.392	0.904	0.866
JPEG	0.05	0.071	0.296	0.972	0.000	0.000	–
	0.10	0.141	0.588	0.959	0.000	0.000	–
	0.20	0.260	0.826	0.918	0.380	0.911	0.875

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Table 12

Historical comparison of smoothed conformal p-values and non-independent CRESS below the target-only floor ($k=4$, all classes, seeds from 0 to 2, five pooled conditions). Randomized rates are exact expectations over the auxiliary randomization. Bold marks $\text{FAR} \leq \alpha+0.02$.

dataset	α	Randomized p-value		CRESS (historical)	
		FAR	power	FAR	power
MVTec	0.05	0.085	0.216	0.049	0.216
	0.10	0.170	0.431	0.105	0.416
VisA	0.05	0.036	0.150	0.051	0.144
	0.10	0.072	0.300	0.095	0.351

Table 13

Historical matched-condition CRESS on 12 label-stratified VisA classes ($k=4$, seeds from 0 to 2, five pooled conditions), using either within-VisA or MVTec source categories. Threshold selection and source assessment are not independent in this historical analysis. Bold marks cells with nonzero power and $\text{FAR} \leq \alpha+0.02$.

source pool	α	FAR	power	prec.	no-harm
VisA (within)	0.05	0.051	0.144	0.847	89%
	0.10	0.095	0.351	0.843	84%
	0.20	0.192	0.637	0.810	78%
MVTec (cross-dataset)	0.05	0.013	0.091	0.936	97%
	0.10	0.040	0.201	0.881	93%
	0.20	0.098	0.415	0.852	93%

Table 14

Protocol-locked representative comparison of fixed LOIO conformal and Vector Platt across cross-dataset, leave-one-category-out (LOCO), and within-split settings. ΔECE is LOIO minus Vector ECE, so negative values favor LOIO; no-harm is the fraction of cells in which LOIO does not increase ECE.

Protocol	Vector ECE	LOIO ECE	ΔECE	No-harm
MVTec $\rightarrow$ VisA	0.301	0.099	-0.203	100%
VisA $\rightarrow$ MVTec	0.304	0.102	-0.202	100%
LOCO	0.304	0.116	-0.188	100%
Within split	0.361	0.154	-0.207	100%

Table 15

MVTec pixel-level localization and FGSM fragility diagnostics, averaged over classes and five seeds. The upper block reports pixel AUROC and AU-PRO; the lower block reports clean and perturbed image AUROC under the label-aware PCA-residual surrogate.

<i>Pixel localization</i>			
k	Method	pixel AUROC	AU-PRO
1	PCA64 (ours)	0.944	0.820
	Controlled DINOv2 NN	0.940	0.845
4	PCA64 (ours)	0.957	0.862
	Controlled DINOv2 NN	0.951	0.862
8	PCA64 (ours)	0.960	0.876
	Controlled DINOv2 NN	0.952	0.863
<i>FGSM fragility diagnostic (image AUROC)</i>			
ϵ	k	clean $\rightarrow$ FGSM	rel. drop
2/255	4	0.937 $\rightarrow$ 0.158	83%
	8	0.945 $\rightarrow$ 0.175	82%
8/255	4	0.937 $\rightarrow$ 0.446	52%
	8	0.945 $\rightarrow$ 0.446	53%

Table 16

Aggregation-sensitivity ablation: clean image AUROC averaged over representative classes and seeds from 0 to 2 for max and top- ρ mean pooling. Bold marks the best aggregator in each row.

dataset	k	max	top-0.5%	top-1%	top-2%	top-5%
MVTec	4	0.9571	0.9638	0.9667	0.9693	0.9621
	8	0.9578	0.9685	0.9740	0.9759	0.9727
VisA	4	0.9039	0.9101	0.9038	0.8944	0.8806
	8	0.9101	0.9200	0.9147	0.9071	0.8941

Table 17

Novelty positioning: each ingredient has established prior art, and this paper’s contribution begins after it. The right column marks where prior work ends and our claims start.

Work family	Established prior art	Our contribution starts after it
AnomalyDINO SubspaceAD	DINOv2 few-shot patch/memory-bank AD DINOv2 PCA/subspace residual ranking	Reliability of the alarms such detectors raise Operational reliability audit downstream of the fixed residual
Calibration/robustness studies	ECE, Platt scaling, FGSM fragility	Conformal-native audit; full calibrator toolbox under one label-free protocol
Conformal AD	P-values from nonconformity scores	The few-shot floor, empirical source-view selection, and finite-category certificate feasibility
Learn-then-Test / CRC / PAC-Wrap	Candidate testing, held-out risk assessment, and one-sided error bounds	Normal-only cross-category $R/P/C$ construction and its independent-category feasibility boundary